Juan Becerra

Software Engineer – Systems, ML Pipelines & Gameplay

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Experience

Electronic Arts – Lifestyle **2025 – Present**
Software Engineer – Tools & Tech-Art *Kirkland, WA*

- Prototyped and integrated ML-assisted authoring and inference pipelines into DCC tools (leveraging Python, PyTorch, and ONNX runtime), enabling automated asset tagging and procedural pose generation.
- Engineered a high-throughput Unreal Engine C++ plugin exporting runtime character trajectory, pose, and velocity feature data to train and populate motion-matching query databases.
- Architected an extensible Maya/Python animation stylization tool and data-processing pipeline adopted across multiple internal studios, reducing redundant manual workflows by over 70%.

Electronic Arts – Maxis Studios **2021 – 2025**
Software Engineer – Gameplay & Animation *Seattle, WA (Remote)*

- Architected a multi-threaded asynchronous path-planning and navigation API in UE5/C++ for *Project Rene* (*The Sims*), enabling deterministic crowd routing and real-time obstacle avoidance within strict frame budgets (<1.5ms).
- Implemented core locomotion and runtime animation systems in UE5 (motion-matching query evaluation, distance matching, custom IK solvers), delivering responsive character controls for simulation gameplay.
- Profiled and optimized runtime memory, compute, and frame budgets across character evaluation graphs and query sweeps, eliminating hitches during dense crowd simulation scenarios.
- Partnered directly with technical animators and gameplay designers to author in-editor tuning tools and visual debuggers, accelerating iteration cycles and runtime diagnostics for complex behavior trees.

Lionbridge **2020 – 2021**
Senior Games Test Engineer *Boise, ID*

- Led a team of QA testers performing combinatorial and regression testing for live-service AAA titles, authoring technical repro steps and engine-level diagnostics in JIRA to accelerate developer triage.

Projects

Hoowan Game Engine | *C++, OpenGL, EnTT, CMake* | GitHub **Nov. 2020**

- Engineered a custom C++ game engine featuring an OpenGL rendering submission pipeline, input event polling, and 2D collision detection and correction systems.
- Integrated an EnTT-based entity-component system (ECS) for data-oriented state evaluation and constructed a companion desktop editor application for rapid scene composition.

DoomSlop | *Three.js, Web Audio API, Vite, JavaScript* | Live Demo | GitHub **Sep. 2026**

- Built a 3D browser first-person shooter featuring hitscan raycasting, synthesized Web Audio sound effects, custom canvas radar minimap, and automated CI/CD deployments via GitHub Pages.

Technical Skills

Languages: Python, SQL, Rust, C++ (Modern, Multi-threading), C#, Java, JavaScript, GLSL/HLSL
Tools & ML Pipelines: PyTorch/ONNX, DCC Pipeline Automation, Unreal Plugin Development, Maya API, PySide/Qt, In-Editor Tooling
Infrastructure & Tools: AWS, Docker, CI/CD, Linux, Git, Perforce (P4), CMake, Visual Studio, JIRA
Gameplay & Systems: Animation Runtime (IK, Motion-Matching), Async Navigation/Pathfinding, Locomotion & Character Physics, State Machines
Game Engines & APIs: Unreal Engine 5 (Source, Motion Matching, GAS, Mass/Crowd), Unity, Godot, Vulkan, OpenGL

Education & Continuing Studies

Boise State University **2017 – 2021**
Bachelor of Science in Computer Science *Boise, ID*

University of Washington (Continuing Education) **2026 – Present**
Graduate Coursework in Machine Learning & Deep Learning (Non-Degree, In Progress) *Seattle, WA*